

Musculoskeletal basis for human movement

人體運動的肌肉骨骼基礎

HHS4867 Functional Movement Science
HHS4867J 功能性運動科學

The movement system: parts working together

動作系統：各部分協同運作



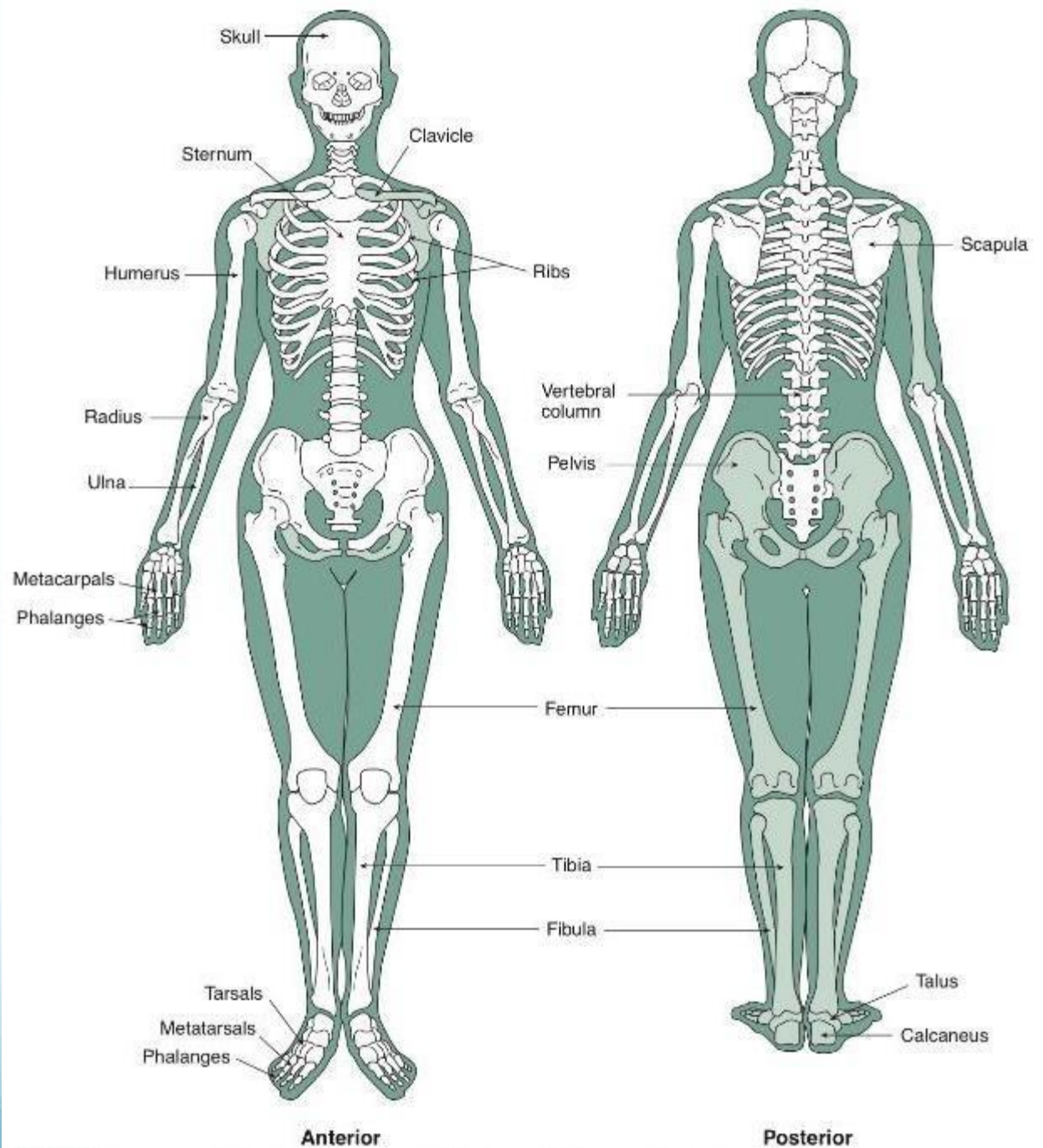
Bones
骨骼

Joints
關節

Muscles
肌肉

Connective tissue
結締組織

Anterior & Posterior View of Human Body 人體前後視圖



Skeleton & Bone Tissue

骨骼和骨組織

- The skeleton is composed of bones, joints, cartilage and ligaments
 - 骨骼由骨骼、關節、軟骨和韌帶組成
- **Joints** are the intersections between bones
 - 關節是骨骼之間的連接點
- **Ligaments** connects bone at the joints, thus reinforcing the joints
 - 韌帶連接關節處的骨骼，從而加強關節

Bone Tissue Function

骨組織功能

● Support

- Provides significant structural support
- Maintain a posture while accommodating large external forces

● Attachment Sites

- Site of attachment for tendons, muscles and ligaments, thus allows for the generation of movement through the muscular system

● Leverage

- Provides the levers and axes of rotation about which the muscular system generates the movement

• 支持

- 提供重要的結構支持
- 適應較大的外力時保持姿勢。

• 附著點

- 肌腱、肌肉和韌帶的附著點，因此可以通過肌肉系統產生運動。

• 槓桿作用

- 提供槓桿和旋轉軸，肌肉系統可以圍繞這些槓桿和旋轉軸產生運動

Cartilage 軟骨

- 2 main types of cartilages:

- **Articular(hyaline) cartilage**

- Covers the articulating ends of the bones in the free moving joints
- Function:
 - Transit compressive forces across the joint
 - Allows motion in the joint with minimal friction
 - Redistributes contact stresses over a larger area
 - Protects underlying bones

- **Fibrocartilage**

- Acts as an intermediary between hyaline cartilage and other connective tissues
- Can withstand high pressures
- Refers to as articular disc, or meniscus

軟骨的2種主要類型：

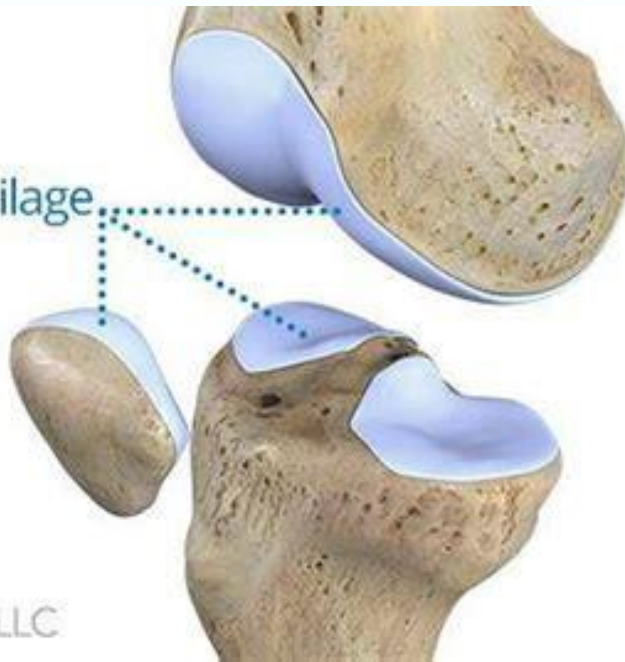
- **關節（透明）軟骨**

- 覆蓋自由活動關節中骨骼嘅關節端
- 功能：
 - 通過關節的傳遞壓縮力
 - 允許關節運動，摩擦最小
 - 在更大的區域上重新分配接觸應力
 - 保護底層骨骼

- **纖維軟骨**

- 充當透明軟骨和其他結締體之間的仲介組織
- 可承受高壓
- 指關節盤或半月板

Articular Cartilage



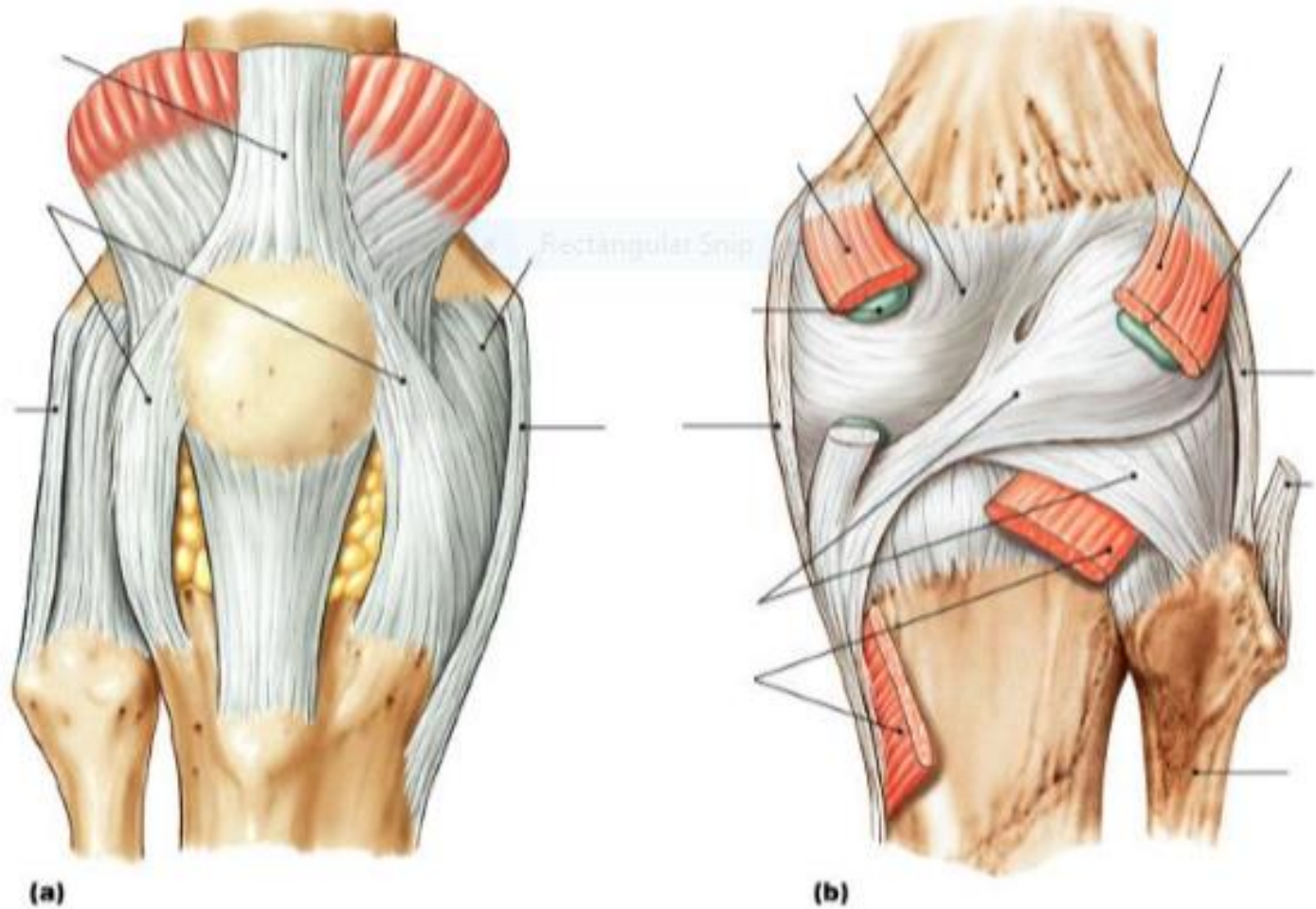
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Meniscus半月板



Ligaments 韌帶

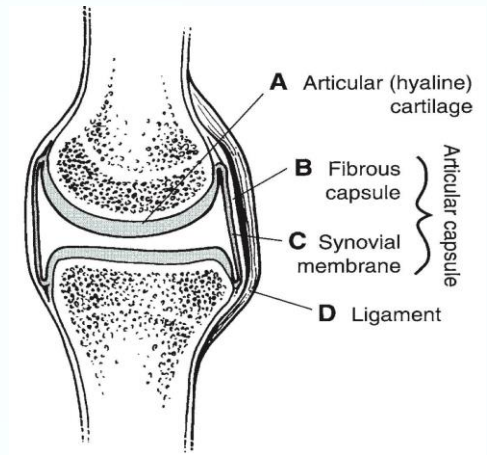
- Short bands of **tough fibrous connective tissue** that binds bones to bones
 - Provides **support in one direction**, and often blends with the capsule of the joint
 - Capsular ligaments: thickenings in the wall of the capsule
 - Extracapsular ligaments: outside the joint
 - Intra-articular ligaments: located inside the joint
 - Function:
 - Guide normal joint movement
 - Restrict abnormal joint movement
- 將骨骼與骨骼結合起來的堅韌纖維結締組織短帶
 - 提供一個方向的支撐，並經常與關節囊融合
 - 包膜韌帶：囊壁增厚
 - 囊外韌帶：關節外
 - 關節內韌帶：位於關節內
 - 功能：
 - 引導關節正常運動
 - 限制關節異常運動



Ligaments: Bone → bone • guides and limits movement
韌帶： 骨與骨連接 • 引導及限制活動

Joint – Synovial Joint

關節 – 滑膜關節



- A – articular cartilage
 - B + C → joint capsule
 - fibrous white connective tissue
 - C: synovial membrane
 - Secretes **synovial fluid** into the joint to lubricate & provide nutrition to the joint
 - A healthy joint provides effortless motion along preferred anatomical directions with an accompanying restriction of abnormal joint motion
- A – 關節軟骨
 - B+C 關節囊
 - 纖維狀白色結締組織
 - C：滑膜
 - 將滑液分泌到關節中，潤滑關節併為關節提供營養
 - 健康的關節可以沿著首選的解剖方向輕鬆運動，並伴隨著異常關節運動的限制

Other Types of Joints

其他類型的關節

● Fibrous Joints

- Little or no movement between the bones & hold the bones together

● Cartilaginous Joints

- Hold bones together with hyaline cartilage or fibrocartilage
- The movement is limited

● 纖維關節

- 骨頭之間很少或沒有運動，並將骨頭固定在一起

● 軟骨關節

- 用透明軟骨或纖維軟骨將骨骼固定在一起
- 運動有限

Types of Joints in the Human Body

人體關節的三種類型

Fibrous Joint

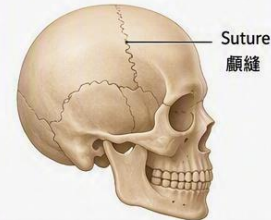
纖維關節

Bones are joined by fibrous connective tissue that allows little or no movement.

骨骼由纖維結締組織連接，幾乎不允許任何運動。

Fibrous Joint

纖維關節



Suture
顱縫

Cartilaginous Joint

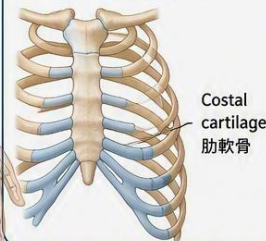
軟骨關節

Bones are joined by cartilage that allows limited movement.

骨骼由軟骨連接，允許有限的運動。

Cartilaginous Joint

軟骨關節



Costal
cartilage
肋軟骨

Synovial Joint

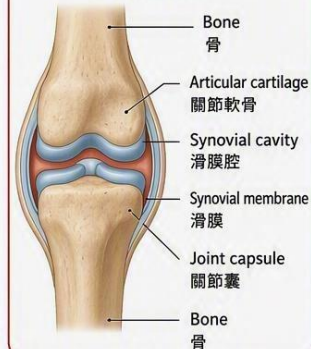
滑膜關節

Bones are joined within a synovial cavity that allows free movement.

骨骼在滑膜腔內連接，允許自由運動。

Synovial Joint

滑膜關節

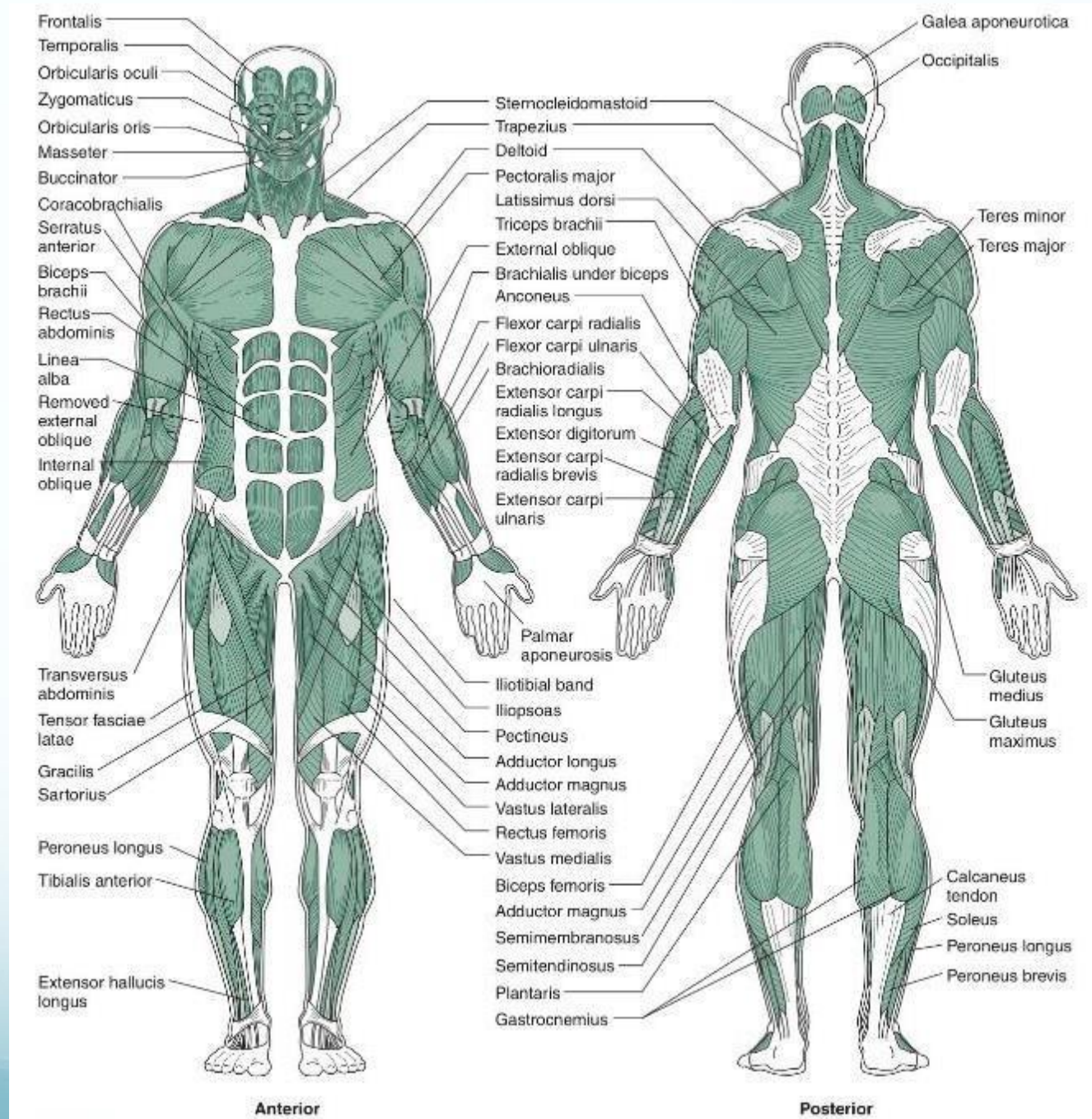


Bone
骨
Articular cartilage
關節軟骨
Synovial cavity
滑膜腔
Synovial membrane
滑膜
Joint capsule
關節囊
Bone
骨

age and synovial fluid.
及滑液。

Skeletal muscles of human body

人體骨骼肌



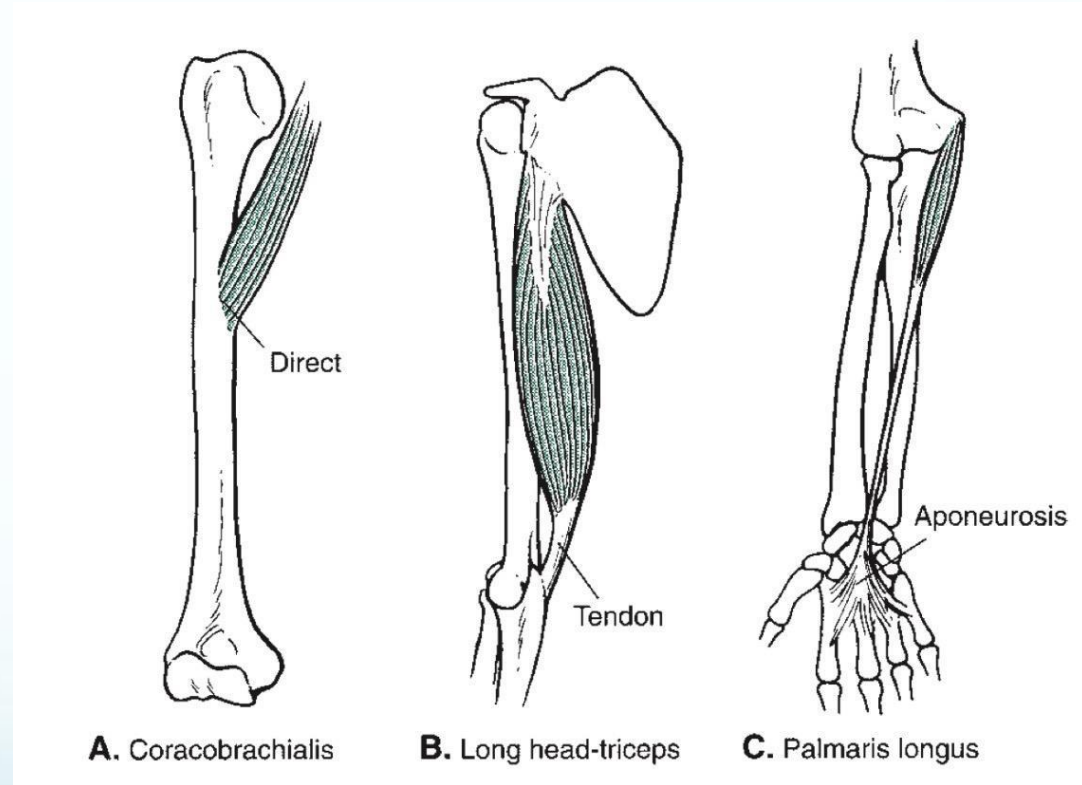
Functions of Skeletal muscle

骨骼肌的功能

- Provide movement
 - Muscle actions generate tensions that are transferred to the bone → create skeletal movement
 - Maintain Postures and Positions
 - Muscle actions of a lesser magnitude
 - Muscle activity is continuous
 - Stabilize Joints
 - Muscle tensions are generated & applied across the joints via tendons, providing stability where they cross the joint
- 提供運動
 - 肌肉動作會產生張力，並轉移到骨骼 → 創建骨骼運動
 - 保持姿勢和姿勢
 - 較小程度的肌肉動作
 - 肌肉活動是連續的
 - 穩定關節
 - 肌肉張力通過肌腱產生並施加到關節上，在它們穿過關節的地方提供穩定性

Transmission of Muscle force to Bone 將肌肉力量傳遞到骨骼

- Muscle attaches to bone in 3 ways
 - Directly into the **bone**
 - Via a **tendon**
 - Via an **aponeurosis**
- 肌肉以 3 種方式附著在骨骼上
 - 直接連接骨頭
 - 通過肌腱
 - 通過腱膜



Role of Muscle

肌肉的作用

- **Origin V.S. Insertion**

- A muscle typically attaches to a bone at both ends
- **Origin:** the attachment closest to the middle of the body, or more proximal
- **Insertion:** the attachment farther from the midline, or more distal; it usually converges to a tendon
- Muscle force is generated & applied to both skeletal connections, resulting in movement of one bone or both

- 起點與止點

- 肌肉通常在兩端都附著到骨骼
- 起點：最靠近身體中線或更近端的附著點
- 止點：離中線較遠或更遠端的附著點；它通常匯聚成一根肌腱
- 肌肉力量產生並施加到兩個骨骼連接處，導致一個或兩個骨骼的運動

Role of Muscle

肌肉的作用

● Developing Torque

- A muscle controls or creates a movement through the development of torque
- **Torque:** the tendency of a force (tension) to produce rotation about a specific axis (the joint)
- The amount of tension is influenced by the **angle of attachment of the muscle**

● Muscle Actions creating, opposing, and stabilizing movements

- In one movement of a joint, e.g. shoulder abduction
- **Agonists:** muscles creating the movement
- **Antagonists:** muscles opposing or producing the opposite joint movement
- **Stabilizers & Neutralizers:** muscles contracting to eliminate the undesired joint action of another muscle

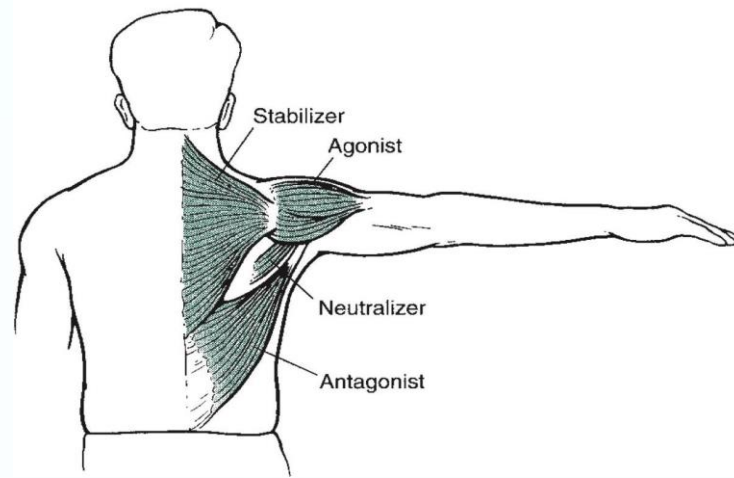
● 產生力矩

- 肌肉通過產生力矩來控制或產生運動。
- 力矩：力（張力）產生繞特定軸（關節）
- 旋轉的趨勢。張力的大小受肌肉附著角度的影響。

● 產生、對抗和穩定運動的肌肉動作。

- 在一個關節的一個運動中，例如肩外展
- 主動肌：產生運動的肌肉
- 拮抗肌：對抗或產生相反關節運動的肌肉
- 穩定肌和中和肌：收縮以消除另一個肌肉的不良關節動作的肌肉

Role of Muscle 肌肉的作用



- In shoulder abduction,
 - **Agonist: Deltoid (abductor)**
 - Abducts the shoulder
 - **Antagonist: Latissimus dorsi (adductor)**
 - Resists the abduction
 - **Stabilizer: Trapezius**
 - Stabilize and hold the scapula in place
 - **Neutralizer: Teres minor**
 - Neutralizes any internal rotation produced by lat. dorsi
- 在肩外展中
 - 主動肌：三角肌（外展肌）
 - 外展肩膀
 - 拮抗肌：背闊肌（內收肌）
 - 抵抗外展
 - 穩定肌：斜方肌
 - 穩定並固定肩胛骨
 - 中和肌：小圓肌
 - 中和背闊肌產生的任何內旋

Net Muscle Actions 淨肌肉動作

- **Isometric** Muscle Action / Contraction

- Muscle tension is generated against resistance to maintain position
- With **no visible change** in joint position

- **Concentric** Muscle Action / Contraction

- Muscle visibly **shortens** while tension is generated

- **Eccentric** Muscle Action / Contraction

- Muscle **lengthens** while tension is generated

- 等長肌肉動作/收縮

- 肌肉張力是針對保持姿勢的阻力產生的
- 關節位置沒有明顯變化

- 向心肌肉動作/收縮

- 肌肉明顯縮短，同時產生肌肉張力

- 離心肌肉動作/收縮

- 肌肉拉長，同時產生肌肉張力

Net Muscle Actions 淨肌肉動作

